SP ENGINEERING ACADEMY

Video Lectures / Course Materials

SUBSCRIBE

COMPLETE COURSE

DIGITAL SYSTEMS

DIGITAL CIRCUIT DESIGN

Lecture Series #15 – Flip Flop

SUBSCRIBE

Like Comments Share

Topic #	Content	Page No.
01	Flip Flop - Definition, Comparison and Explanation - https://youtu.be/dnlkkpwwA2U	02
02	Positive Edge Triggered SR Flip Flop - Truth Table, Excitation Table, Logic Circuit and Explanation - https://youtu.be/GBvGaBUFbks	05
03	Negative Edge Triggered SR Flip Flop - Truth Table, Excitation Table, Logic Circuit and Explanation - https://youtu.be/-BINwZqjk9Y	07
04	D Flip Flop - Truth Table, Excitation Table, Logic Circuit and Explanation - https://youtu.be/dlulHgnnCR0	08
05	JK Flip Flop - Truth Table, Excitation Table, Logic Circuit and Explanation - https://youtu.be/zYjiTxsQpz0	10
06	T Flip Flop - Truth Table, Excitation Table, Logic Circuit and Explanation - https://youtu.be/eb-G7AgNgOw	12
07	Race Around Condition - Flip Flop - https://youtu.be/Q5PAHFyS2zc	---
08	Master Slave SR Flip Flop - Truth Table, Logic Circuit and Explanation - https://youtu.be/0H5tYsm05mw	14
09	Master Slave JK Flip Flop - Truth Table, Logic Circuit and Explanation - https://youtu.be/YzdUfjnPMB4	16

SCSA1201 - Fundamentals of Digital Systems

Unit 4 - Synchronous Sequential Logic

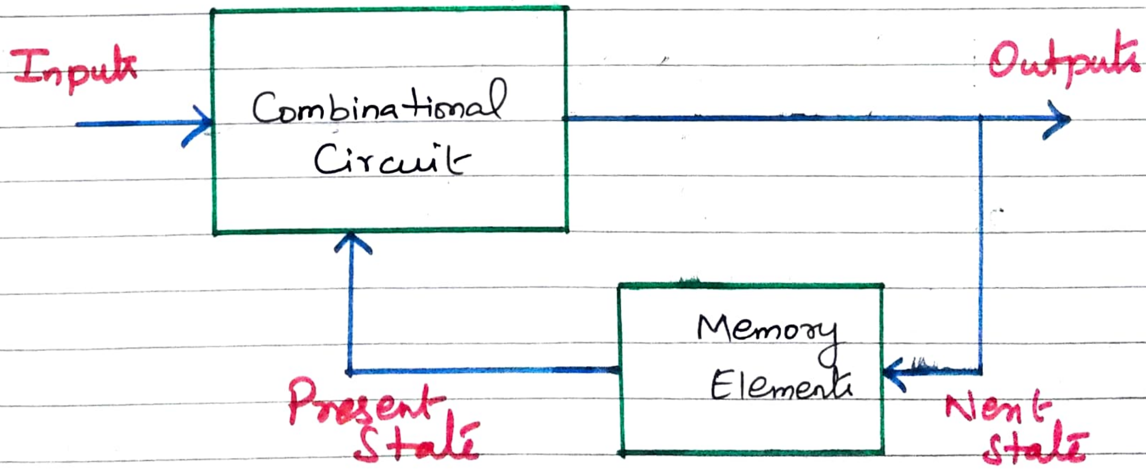
Flip Flops - Analysis of clocked Sequential Circuit - Reduction and assignments - Flip Flop excitation tables - Design Procedure - Design of Counters - Registers - Shift registers - Synchronous Counters - Timing Sequences - Algorithmic State Machines - ASM chart - timing Considerations - Control implementation

Video Link - https://youtu.be/dnlkkpwwA2U

Sequential Circuit / Finite State Machine (FSM)

Diagram shows the block diagram of Sequential circuit / Finite State Machine (FSM).

Memory elements are Connected to the Combinational circuit as a feedback path



→ The information stored in the memory elements at any given time defines the present state of the sequential circuit

→ The present state and the external inputs determines the outputs and the next state of the Sequential Circuit

→ Thus we can specify the Sequential Circuit by a time sequence of external inputs, internal states (present state and next states) and outputs

→ The Counters and registers are the Common examples of Sequential Circuits

The memory elements used in Sequential Circuits is a flip flop which is Capable of Storing 1-bit binary information

Comparison between Combinational and Sequential Circuits

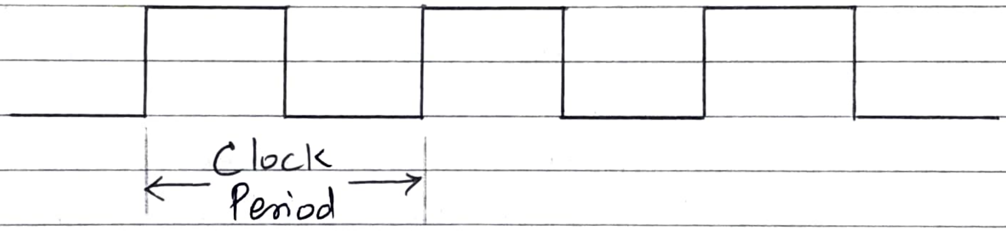
S. No.	Combinational Circuits	Sequential Circuits
1.	In Combinational Circuits, the output Variables are at all times dependent on the Combination of input Variables	In sequential Circuits, the output Variables depend not only on the present input Variables but they also depend upon the past history of these input Variables
2.	Memory unit is not required,	Memory unit is required to Store the past history of inputs Variables
3.	Combinational Circuits are faster in Speed because the delay between input and output is due to propagation delay of gates	Sequential Circuits are Slower than the Combinational Circuits
4.	Easy to design	Harder to design
5.	Eg - Parallel Adder	Eg - Serial adder

Clock

→ A clock signal is a particular type of signal that oscillates between a high and a low state and is utilized to coordinate actions of circuits

→ It is produced by a clock generator

→ The most common clock signal is in the form of a square wave with a 50% duty cycle, usually with a fixed, constant frequency as shown in diagram



→ Circuits using the clock signal for synchronization may become active at either the rising edge, falling edge or in the case of double data rate, both in the rising and in the falling edges of the clock cycle

→ The time required to complete one cycle is called clock period or clock cycle

→ Ideally, the clock signal should have sharp transitions from one level to other

Flip-Flop

In electronics, a flip-flop or latch is a circuit that has two stable states and can be used to store state information. Flip-flops and latches are used as data storage elements. A flip-flops stores a single bit (binary digit) of data; one of its two states represents a "one" and other represents a 'zero'.

SR Flip-Flop

→ The SR flip-flop also known as a SR latch, Can be Considered as one of the most basic Sequential logic Circuit possible

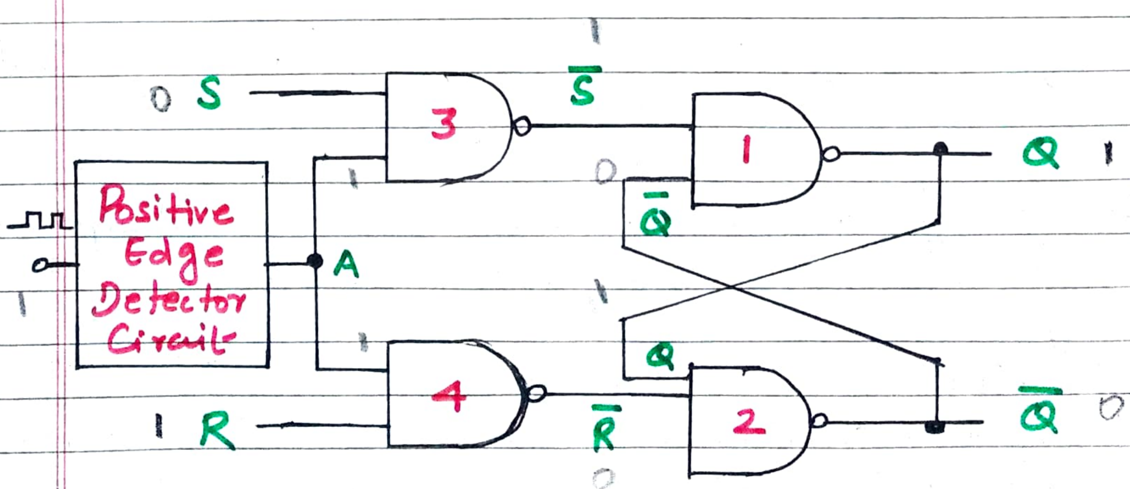
→ This simple flip-flop is basically a one bit memory bistable device that has two inputs, one which will 'SET' the device (meaning the OUTPUT = '1'), and is labelled S and another which will "RESET" the device (meaning the output = "0"), labelled R.

→ Then the SR description Stands for "set - Reset". The Set input resets the flip flop back to it's original State with an output Q that will be either at a logic level "1" or logic "0" depending upon this set / reset Condition.

Video Link - https://youtu.be/GBvGaBUFbks

(a) Positive Edge Triggered SR Flip Flop

The diagram Shows the positive edge triggered clocked SR flip flop.



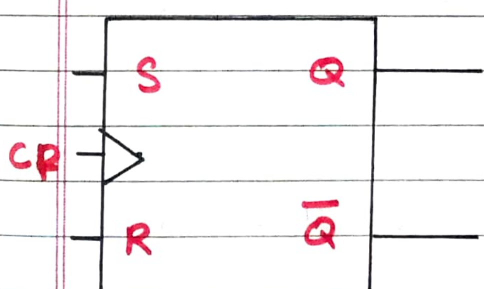
→ The Circuit is Similar to SR latch except enable signal is ~~xp~~ replaced by the Clock Pulse (CP) followed by the positive edge detector Circuit

→ The edge detector Circuit is a differentiator

→ The diagram Shows input and output waveforms for positive edge triggered clocked SR flip-flop

→ The Circuit output responds to the S and R inputs only at the positive edges of the clock pulse. At any other instants of time, the SR flip-flop will not respond to the changes in input

Logic Symbol



CP - Clock Pulse

Truth Table

CP	S	R	Q_n	Q_{n+1}	State
↑	0	0	0	0	No Change (NC)
↑	0	0	1	1	
↑	0	1	0	0	Reset
↑	0	1	1	0	
↑	1	0	0	1	Set
↑	1	0	1	1	
↑	1	1	0	X	Intermediate
↑	1	1	1	X	
0	X	X	0	0	No Change (NC)
0	X	X	1	1	

Excitation Table

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

Truth Table

S	R	Q_{n+1}
0	0	Q_n
1	0	1
0	1	0
1	1	-

Characteristic Equation

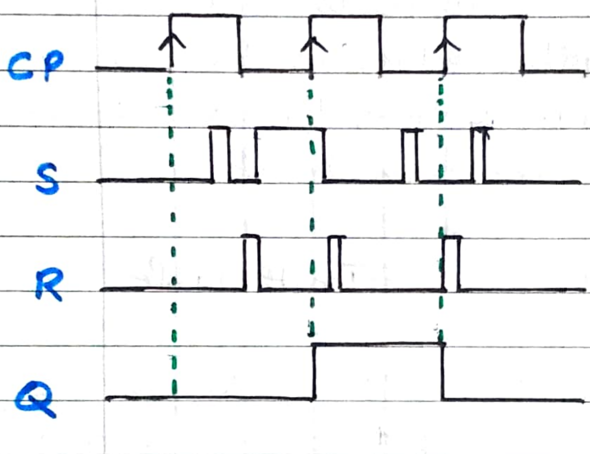
SR \ Q_n	0	1
00	0	1
01	0	0
11	X	X
10	1	1

$Q_{n+1} = S + \overline{R}Q_n$

Case 1 – If $S = R = 0$ and the clock pulse is applied, the output do not change, (ie)
$Q_{n+1} = Q_n$.

Case 2 – If $S=0$, $R=1$ and the clock pulse is applied,
$Q_{n+1} = 0$

Case 3 – If $S=1$, $R=0$ and the clock pulse is applied, the State of the flip-flop is Undefined

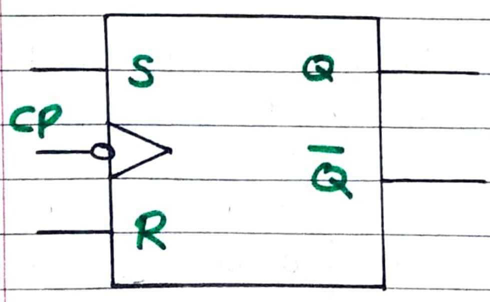


Video Link - https://youtu.be/-BINwZqjk9Y

(b) Negative Edge Triggered SR Flip-Flop

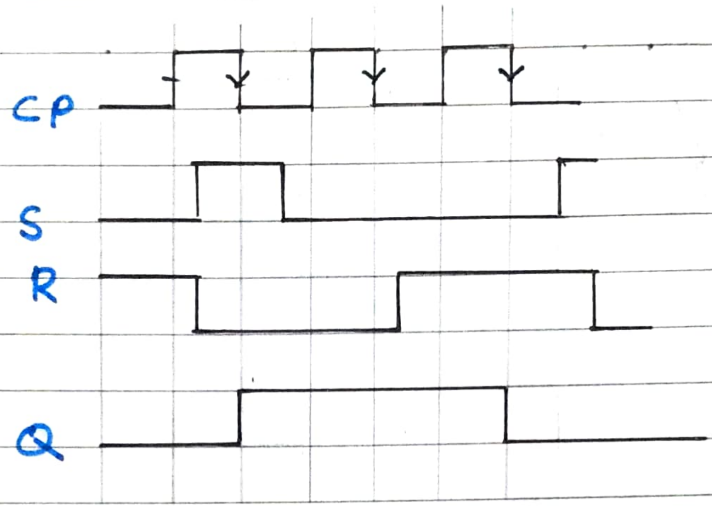
In the negative edge triggered SR flip-flop, the negative edge detector Circuit is Used and the Circuit output responds at the negative edges of the clock pulse

Logic Symbol



CP – Clock Pulse

CP	S	R	Q_n	Q_{n+1}	State
↓	0	0	0	0	No
↓	0	0	1	1	Change (NC)
↓	0	1	0	0	Reset
↓	0	1	1	0	
↓	1	0	0	1	Set
↓	1	0	1	1	
Ø ↓	1	1	0	×	Intermi
Ø ↓	1	1	1	×	nate



D Flip Flop

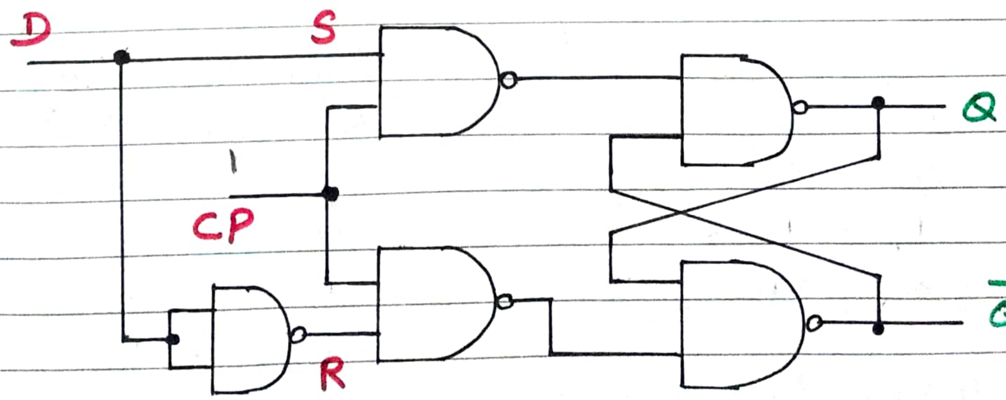
Video Link - https://youtu.be/dIulHgnnCR0

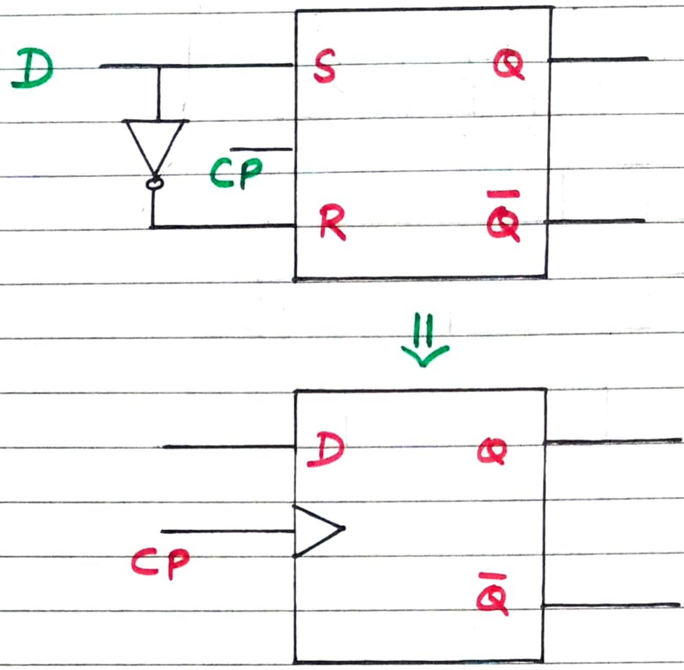
→ The basic building block of D flip flop is a SR flip flop

→ The SR flip flop has two data inputs S and R

→ The S input is made high to store 1 in the flip-flop and R input is made high to store 0 in the flip flop

→ When both inputs are same the output either does not change or it is invalid (Inputs → 00, no change and inputs → 11, invalid)





→ In many applications, these inputs Conditions are not required

→ These input Conditions can be avoided by making them Complement of each other. This modified SR flip flop is known as D flip flop

→ The D input goes directly to the S input, and its Complement is applied to the R input. Due to these Conditions, only two input Conditions exists, either $S=0$ & $R=1$ or $S=1$ & $R=0$

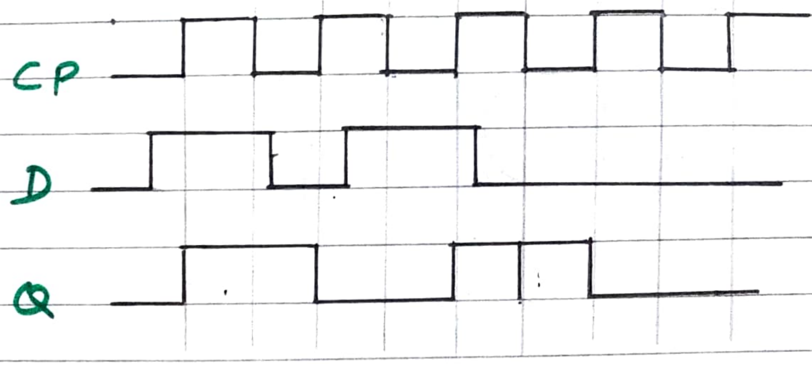
Truth Table

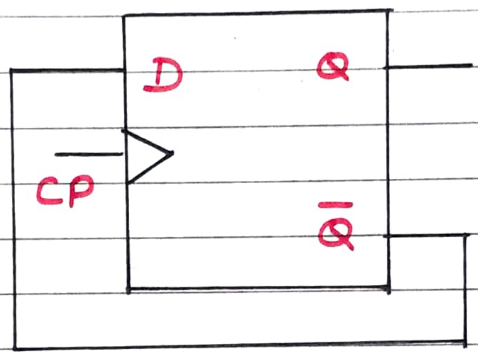
D	Q_{n+1}
0	0
1	1
X	Q_n

Excitation Table

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

D	Q_n	Q_{n+1}
0	0	0
0	1	0
1	0	1
1	1	1





If we Connect the $\bar{Q}$ output of D flip-flop to its D input as Shown in diagram, the output of D flip flop will change either from 0 to 1, or from 1 to 0 at every positive edge of the D flip flop

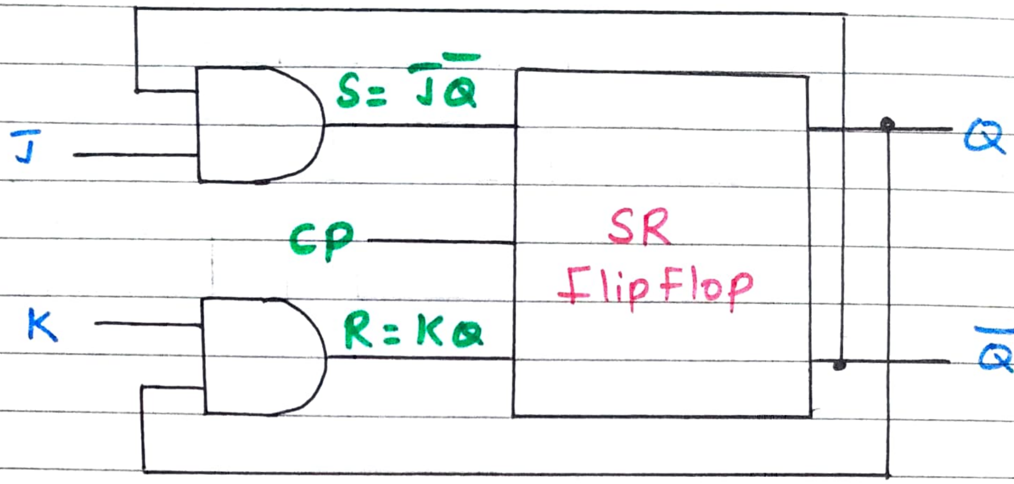
Such change in the output is Known as toggling of the flip-flop output

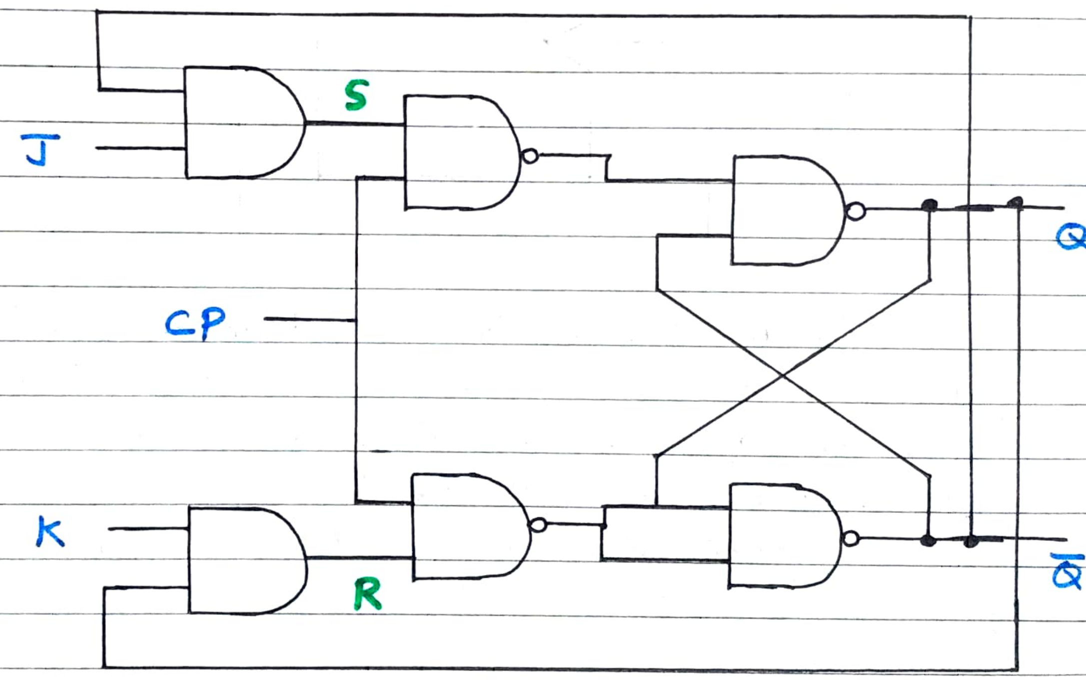
JK Flip Flop

Video Link - https://youtu.be/zYjiTxsQpz0

→ The uncertanity in the State of an SR flip flop when $S = R = 1$ Can be eliminated by Converting it into a JK flip flop

→ The data inputs are J and K which are ANDed with Q and $\bar{Q}$ respectively to obtain S and R inputs, as Shown in diagram. $S = J \cdot \bar{Q}$ and $R = K \cdot Q$





Truth Table

Q_n	J	K	Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

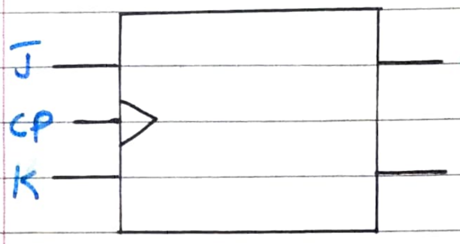
$\Rightarrow$

J	K	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	$\overline{Q_n}$

Excitation Table

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

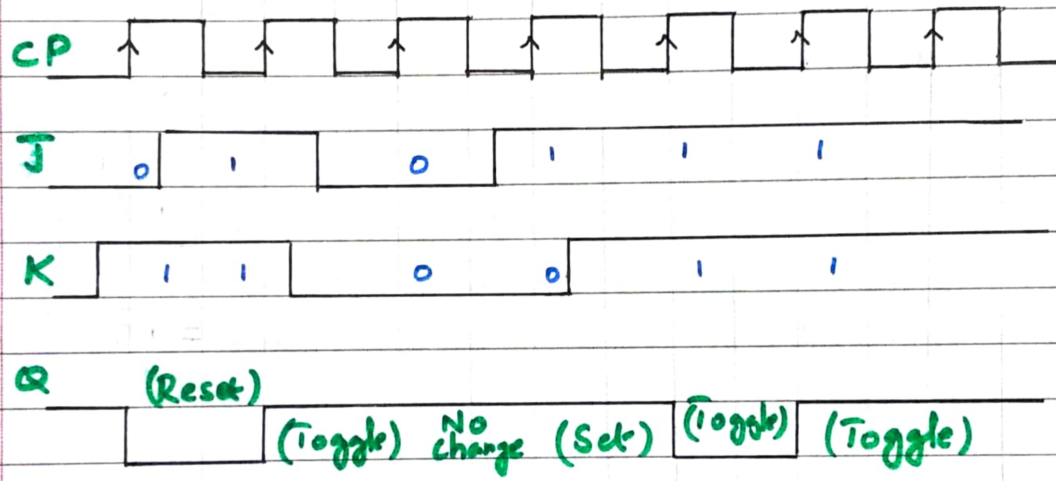
Logic Symbol



Characteristic Equation

Q_n \ JK	00	01	11	10
0	0	0	1	1
1	1	0	0	1

$$Q_{n+1} = \overline{Q}_n J + Q_n \overline{K} +$$



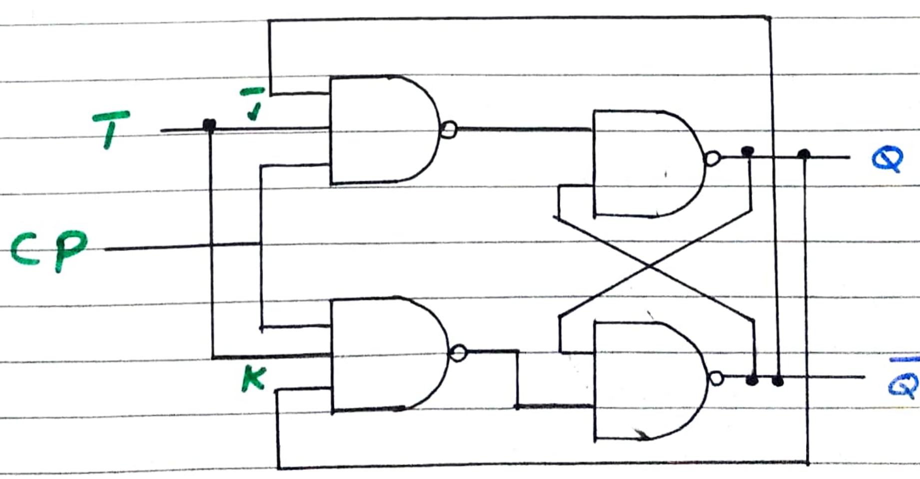
Video Link - https://youtu.be/eb-G7AgNgOw

T Flip Flop

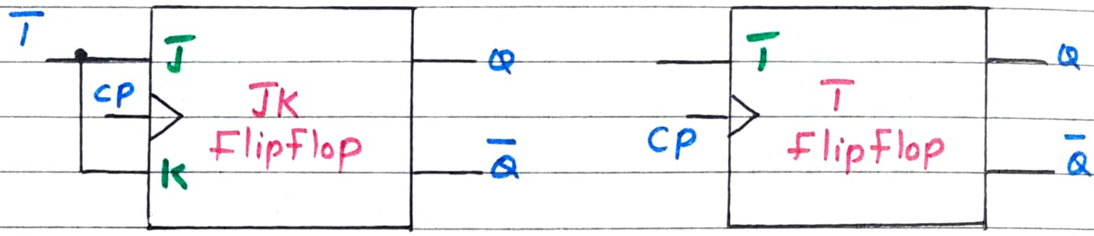
→ T flip flop is known as Toggle flip flop

→ The T flip flop is a modification of JK flip flop

→ T flip flop is obtained from a JK flip flop by connecting both inputs J and K together



Logic Symbol



Characteristic Equation

Q_n \ T	0	1
0	0	1
1	1	0

$$Q_{n+1} = T\overline{Q_n} + \overline{T}Q_n$$

Truth Table

Q_n	T	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

T	Q_{n+1}
0	Q_n
1	$\overline{Q_n}$

Excitation Table

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

T	Q_n	Q_{n+1}	
0	0	0	No change
0	1	1	
1	0	1	Toggle
1	1	0	

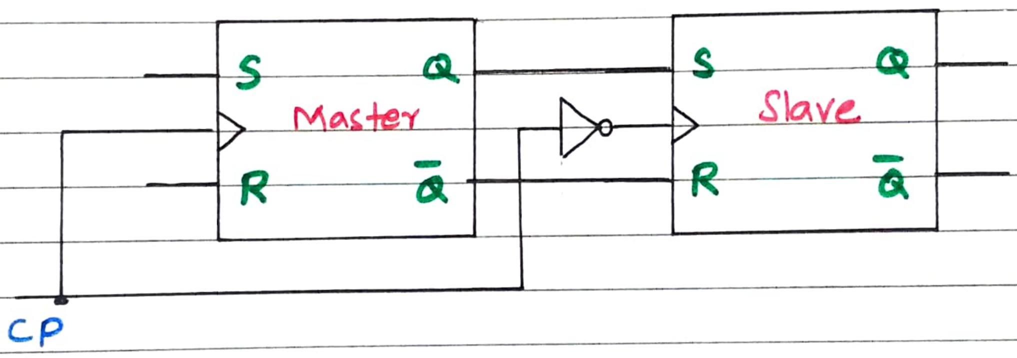
Master Slave SR Flip Flop

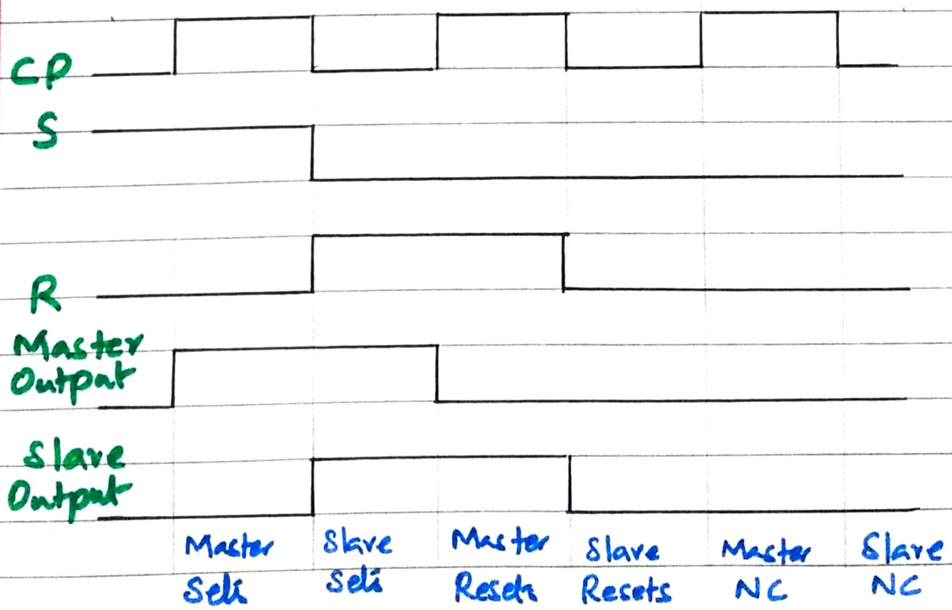
Video Link - https://youtu.be/0H5tYsm05mw

→ A master slave flip flop is Constructed from two flip flops

→ One Circuit Serves as a master and the other as a Slave, and the overall Circuit is referred as a master-Slave flip flop

→ Diagram Shows SR master-Slave flip flop





→ It Consists of a master flip-flop, a Slave flip-flop and a inverter

→ Both the flip-flops are positive level triggered, but inverter Connected at the clock input of the slave flip flops forces it to trigger at the negative level

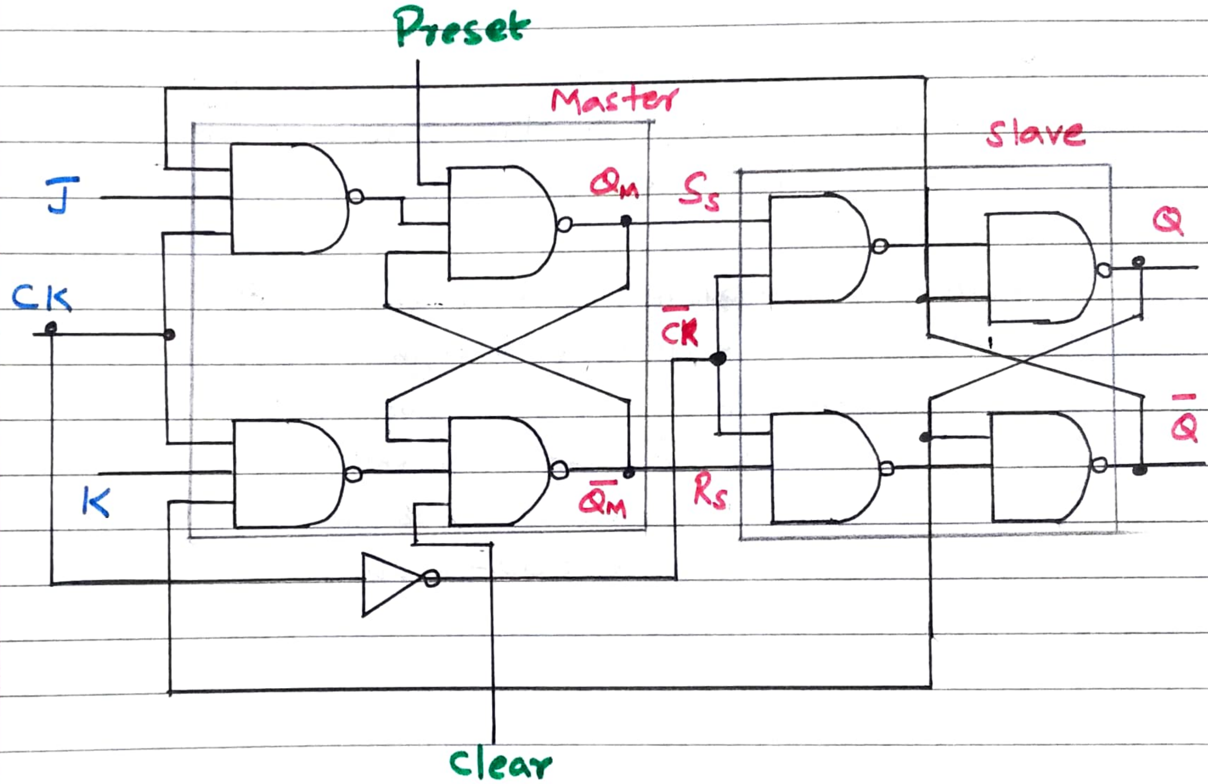
→ The output State of the master flip-flop is determined by the S and R inputs at the positive clock pulse

→ The output State of the master is then transferred as an input to the Slave flip-flop. The Slave flip-flop uses this input at the negative clock pulse to determine its output State

Master Slave JK Flip-Flop

Video Link - https://youtu.be/YzdUfjnPMB4

→ Diagram shows the master-slave JK flip-flop. Positive clock pulses are applied to first flip-flop and inverted clock pulses are applied to second flip-flop



→ when CK = 1, the first flip-flop is enabled and the outputs Q_M and $\overline{Q_M}$ responds to the inputs of J and K according to the table. At this time, the second flip flop is inhibited because its clock is low, $\overline{CK} = 0$

Truth Table

Q_n	J	K	Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

J	K	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	$\overline{Q_n}$

→ when CK goes low ($\overline{CK} = 1$), the first flip-flop is inhibited and second flip-flop is enabled. At this time, the output of second flip-flop (Q and $\overline{Q}$) follow the outputs (Q_M and $\overline{Q}_M$) respectively

→ Since the second flip-flop follows the first one, it is referred to as the slave and the first one as the master

→ In master-slave JK flip-flop state change occurs when flip-flop goes through both positive transition (first half) of clock and negative transition of the clock (second half). Thus, race-around condition does not exists in the master-slave JK flip-flop